

Retail Prices Analysis and Machine Learning Prediction Report

- **Git Repo Link :** https://github.com/AdditionalBug/Numarical_Maths.git

1.Introduction

Project Objective

This project is to analyze retail market prices, perform statistical hypothesis testing, machine learning models to accurately forecast average retail prices based on temporal, categorical, and price range features using exsisting retailprice dataset.

Dataset Description

The dataset contain weekly retail market pricing data across multiple grocery and agricultural items. It includes average prices, low prices, and high prices across different units of measurement.

Number of Records and Variables

- **Rows/Records:** 22,449
- **Columns/Variables:** 6
 1. Week
 2. Item
 3. Unit
 4. Avg Price (Rs.)
 5. Price Range – Low Price (Rs.)
 6. Price Range – Hiigh Price (Rs.)
- **Link :** <https://www.kaggle.com/datasets/chamodperera87/weekly-retail-prices-in-colombo-sri-lanka>

- **Dataset Name:** Weekly Retail Prices in Colombo, Sri Lanka (retail_prices.csv)

2.Methodology

Data Cleaning

- **Cleaning Price Columns:** Text and symbols instead of numbers that was in price columns were removed. Then the values in columns converted to floating values.
- **Missing Value Imputation:** Rows containing missing values in Avg Price (Rs.) = (Target value) were dropped. Missing values in low and high price ranges were filled with the recorded average price for each row.
- **Temporal Encoding:** Converted week labels like (2021-05-W1) in to sequential numbers like (w1, w2, ...) and numbers (Week_Num) to use for machine learning models.

Data Preprocessing

- **Encoding Categorical Variables:** Categorical text features (Item and Unit) were transformed into numerical format using label encoder.
- **Feature Engineering:** A new feature Price_Spread was created by subtracting the low price from the high price range to measure market price gap.
- **Feature Scaling:** Scaled input features using StandardScaler to get them on same numerical scale.
- **Train Test Data Splitting:** The dataset was split into training (80%) and testing (20%) sets using train test split with a fixed random seed (random state = 42) for reproducibility.

Data Visualization

1. **Distribution Histogram:** Overall skewness and frequency distribution of average retail prices.
2. **Box Plot:** Compared price variation for the top 5 most frequently recorded items.
3. **Scatter Plot:** Linear relationship between low price ranges and high price ranges.
4. **Correlation Heatmap:** Check how much features relate to each other.

Hypothesis Testing

Two statistical tests were conducted:

1. **Independent Two Sample Welch's t-Test:** Tested whether significant price differences exist between distinct items.
2. **Pearson Correlation Test:** Tested whether retail prices exhibit a statistically significant linear trend over successive weeks (Week_Num).

Machine Learning Models

Two regression models were trained and evaluated to predict Avg Price (Rs.):

1. **Linear Regression:** A simple baseline model.
2. **Random Forest Regressor:** An advanced model build using decision trees($n_estimators=50$) designed to capture data patterns.

3. Results and Discussion

Key Findings from Visualizations

- **Price Distribution:** The average price distribution is below Rs. 500, but some items (specialized meats or imported goods) go past Rs. 3,000.
- **Item price variation:** Staple items stay within stable price ranges, where premium (Broiler) shows wider price changes.
- **Price Ranges:** The low and High prices follow closer to each. When one goes up other one follows considerably.
- **Correlation Structure:** The correlation heatmap shows high correlation between Avg Price and Price difference. (High price – Low price), while Price_Spread also showed a strong positive correlation ($r = 0.81$) with average prices .

Hypothesis Testing Results

- **Test 1 (Welch's t-Test):**
 - H_0 : There is no significant difference in mean price between items.
 - H_1 : There is a significant difference in mean price.
 - **p-value:** $p = 6.67 \times 10^{-66} < 0.05$
- **Conclusion:** Reject H_0 . There is definitive statistical evidence that different retail items command significantly distinct price points.

- **Test 2 (Pearson Correlation Test):**
- H0: There is no significant difference in mean price between items.
- H1: There is a significant difference in mean price.
- **p-value:** $p = 6.67 \times 10^{-66} < 0.05$
- **Conclusion:** Reject H0. There is definitive statistical evidence that different retail items command significantly distinct price points.

Model Evaluation Results

- **Linear Regression:** Achieved an RMSE of **61.50** and an R^2 score of **0.9939**, showing strong predictive capability.
- **Random Forest Regressor:** Performing better with a lower RMSE of **49.18** and a higher R^2 score of **0.9961**, while capturing more complex patterns in the data.

4. Conclusion and Recommendations

Summary of Findings

This study analyzed over 22,000 retail pricing records. Statistical tests showed clear price differences between items and a steady price increase over time. The machine learning models successfully predicted prices with high accuracy.

Recommendations

1. **Policy Monitoring:** Regulatory bodies can utilize the developed Random Forest predictive framework to flag abnormal price spikes or excessive price spreads (Price_Spread) across local vendors.
2. **Feature Expansion:** Future iterations of this research should integrate external macroeconomic indicators such as fuel prices, seasonal weather indices, and transport costs to further improve predictive granularity.

5. References

1. Weekly Retail Prices in Colombo, Sri Lanka (retail_prices.csv). <https://www.kaggle.com/datasets/chamodperera87/weekly-retail-prices-in-colombo-sri-lanka>

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3. McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*, 51-56.